

Dynamical Structure-Function Correlations of fMRI Human Brain Signals under LSD

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Abstract

Psychedelics like LSD can dramatically reshape our inner experience, yet exactly how it reconfigures brain activity is still unclear. To investigate, we adapted a validated k-means clustering approach—previously applied to sleep, anesthesia, and meditation—to analyze dynamic functional connectivity in fMRI scans of 16 adults under both LSD and placebo.

Our objectives were to (1) quantify how LSD shifts the prevalence of five canonical phase-locking states, (2) assess whether it alters the unpredictability of transitions between those states, and (3) visualize large-scale network reorganization via chord diagrams. We measured fractional occupancy, computed Structure–Function Correlation scores, calculated Shannon and Markov-chain entropies, and generated chord diagrams of state-specific coherence.

Key results show that LSD significantly reduces time in the most anatomy-bound state (State 1) and boosts occupancy of a mid-alignment state (State 4), while leaving overall transition entropy unchanged. Chord diagrams further reveal LSD’s dual effect of enhancing global integration in some states and focusing connectivity in others.

These findings demonstrate that LSD reshapes the brain’s dynamic repertoire, rebalancing which states dominate without increasing random switching, offering mechanistic insights into altered consciousness and potential biomarkers for guiding psychedelic-assisted therapies.

2 Introduction

2.1 State of the Art

Over the past decade, researchers have shown that the brain’s shifting connectivity patterns hold the key to understanding consciousness. Barttfeld et al. demonstrated that certain dominant phase-locking configurations reliably distinguish wakefulness from anesthesia in humans (Barttfeld et al., 2015). Uhrig et al. then dug deeper into resting-state dynamics under anesthesia, revealing similar signatures of consciousness loss in controlled settings (Uhrig et al., 2018). Demertzi et al. took it a step further by showing that these time-varying coordination patterns break down in patients with disorders of consciousness, linking dynamic complexity directly to awareness (Demertzi et al., 2019).

Most recently, Castro et al. brought these strands together with a LEiDA-based clustering pipeline, identifying five robust brain states that serve as generalizable markers of consciousness across sleep and anesthesia (Castro et al., 2024).

In this project, we push this trajectory one step further by applying the same validated pipeline to psychedelic-induced states—specifically LSD. From this approach, we open a new window onto how consciousness can be dynamically reconfigured by psychedelics.

2.2 Research Question

To what extent does LSD alter the Dynamical Structure of the whole-brain connectivity interacting in an altered conscious experience?

Psychedelics such as LSD are known to induce profound changes in subjective awareness, yet the precise ways in which they reshape the dynamic landscape of conscious brain states remain poorly understood. By applying dynamic functional connectivity analysis and k-means clustering to fMRI data, we track how LSD alters both the prevalence and stability of core network configurations. Examining state transition probability matrices, we can uncover whether LSD biases the brain toward particular switching patterns or increases overall unpredictability.

Together, these data driven analyses illuminate both the macro-scale reorganization of the conscious repertoire and the micro-scale pathways by which the brain transitions through altered states.

2.3 Relevance

Understanding how LSD reshapes the repertoire and transitions of whole-brain network states is crucial for several reasons:

- **Mechanistic insights into consciousness:** By linking specific defined states to altered subjective experiences (e.g. ego dissolution, heightened sensory awareness), we advance theories of how dynamic neural configurations give rise to conscious content and its modulation by psychedelics.
- **Therapeutic biomarkers:** Psychedelic-assisted therapies show promise for treatment-resistant conditions (depression, PTSD). Dynamical metrics such as state occupancy and transition flexibility may serve as objective biomarkers to predict and monitor treatment response.
- **Methodological generalizability:** Demonstrating that a validated five-cluster LEiDA pipeline—originally applied across meditation and other altered states—robustly captures psychedelic dynamics underscores its utility for cross-study integration and reproducibility in consciousness research (Panitz et al., 2024; Castro et al., 2024).

3 Data

3.1 Description and Experimental Design

The dataset used here is identical to that of Kaelen et al. (2016), which comprises functional fMRI recordings from $N = 16$ healthy adult volunteers who each completed two single-blind experimental sessions under LSD (75 μ g, intravenous) and placebo conditions.

For each run, whole-brain volumes were acquired ($TR \approx 2$ s, $TE \approx 30$ ms), yielding approximately 225 timepoints per scan. Data were preprocessed following standard pipelines (motion correction, spatial smoothing, temporal filtering) and parcellated into 90 regions of interest.

In this project, we leveraged these preprocessed time series to compute LEiDA centroids and associated metrics (centroid probability, consecutive run lengths, transition probabilities, Markov entropy, and inter-state correlation chord diagrams), allowing a visualization-driven examination of how LSD alters the flow of conscious brain states.

3.2 Processing Techniques

Before applying the clustering algorithm, we applied a series of processing steps to ensure that our BOLD signals accurately reflect neural timing relationships and are directly comparable across regions.

3.2.1 Temporal Standardization

First, each region's raw BOLD time series $x_i(t)$ was normalized via z-score,

$$z_i(t) = \frac{x_i(t) - \mu_i}{\sigma_i},$$

where μ_i and σ_i are that region's mean and standard deviation.

This step puts all regions on the same scale, preventing any one signal's amplitude range from dominating the analysis.

3.2.2 Hilbert Transform and Phase Extraction

From each standardized series $z_i(t)$, we computed the analytic signal via the Hilbert transform:

$$a_i(t) = z_i(t) + j \mathcal{H}\{z_i(t)\}.$$

The instantaneous phase

$$\phi_i(t) = \arg(a_i(t))$$

tells us exactly “where” in its oscillation cycle the signal is at each moment. We focused on phase because it captures timing relationships between regions without being influenced by overall power fluctuations.

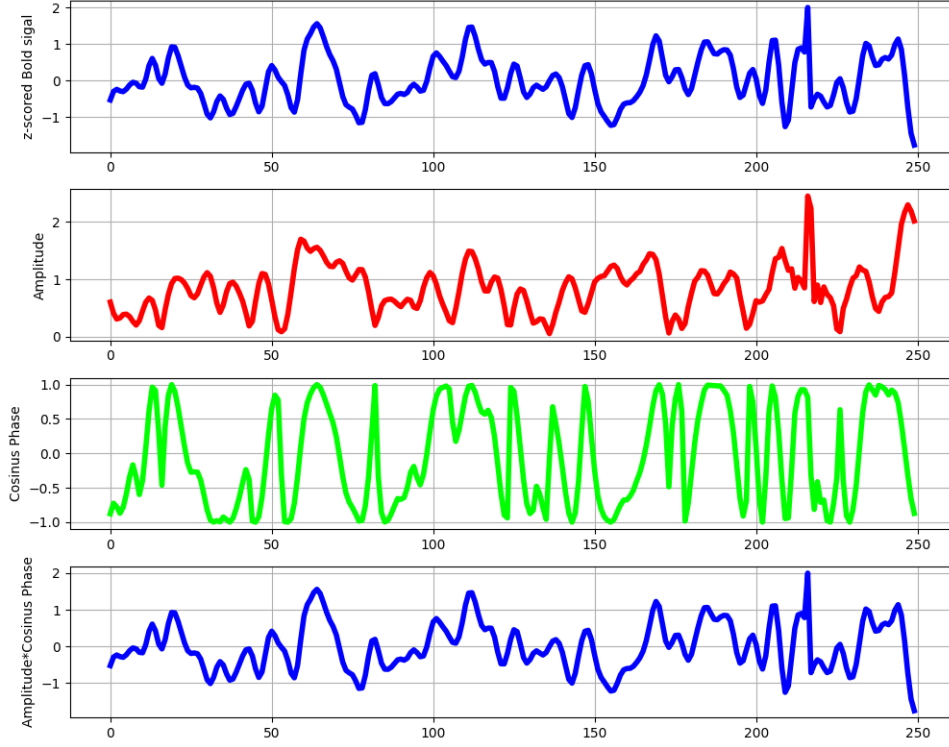


Figure 1: Decomposing a single ROI's BOLD signal with the Hilbert transform. Only the phase $\phi_i(t)$ is used for coherence analysis.

- (a) Z-scored BOLD;
- (b) Instantaneous amplitude $|a_i(t)|$;
- (c) Phase cosine $\cos(\phi_i(t))$;
- (d) Amplitude-modulated phase $A_i(t) \cos(\phi_i(t))$.

3.2.3 Phase Coherence Calculation

At each time point t , we measured synchrony between regions i and j as

$$C_{ij}(t) = \cos(\phi_i(t) - \phi_j(t)),$$

which ranges from 1 (perfect synchrony) to -1 (perfect anti-phase), as shown in Figure 2. This yields a full $N \times N$ coherence matrix at every t .

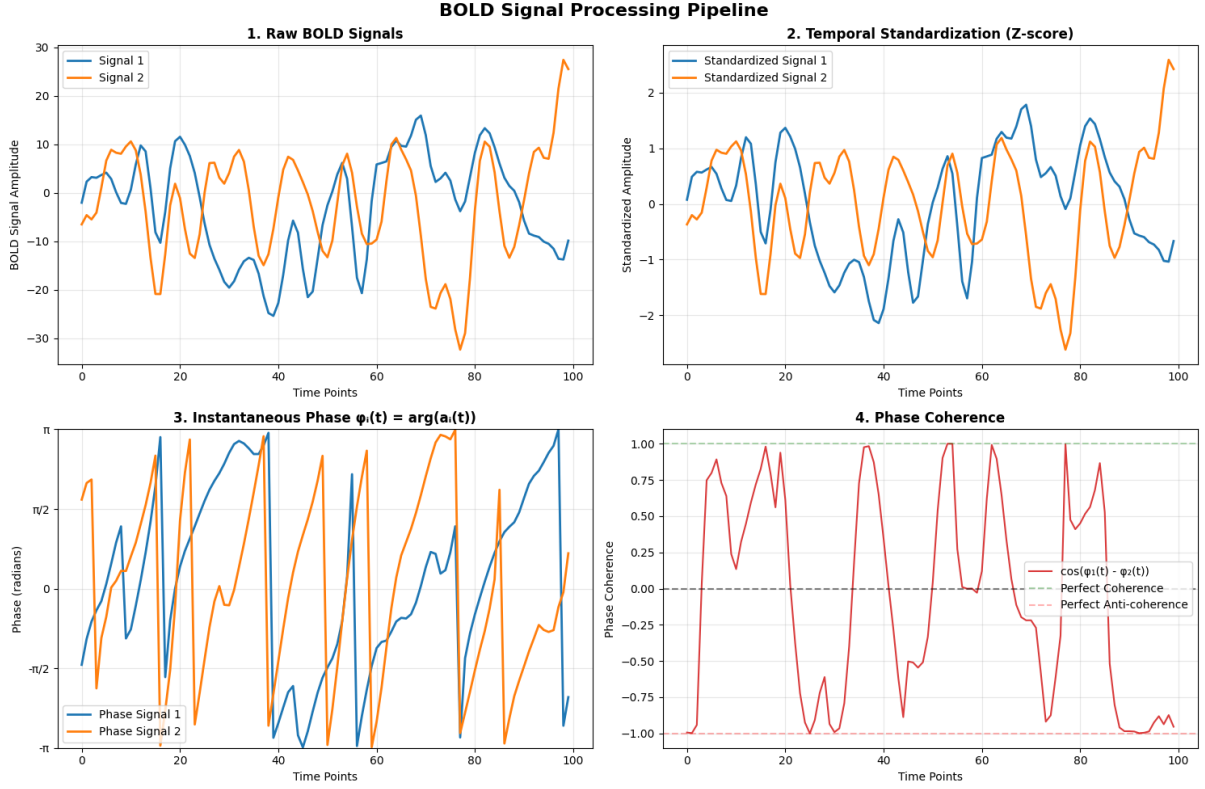


Figure 2: **Complete BOLD Processing Pipeline.**

- (1) Two example raw BOLD signals,
- (2) after z-score standardization,
- (3) instantaneous phases from the Hilbert transform,
- (4) phase coherence at each time point ($\cos(\phi_1 - \phi_2)$).

These coherence values are then vectorized for clustering. Pipeline validated in (Demertzi et al., 2019) and (Castro et al., 2024).

3.2.4 Vectorization for Clustering

To prepare for k-means, each symmetric coherence matrix was reduced to a 1D feature vector by extracting its lower triangular entries (excluding the diagonal). This results produces $\frac{N(N-1)}{2}$ features per time point. For our $N = 90$ regions, that means 4,005 features at each of the $T = 6,937$ time points, resulting in a $6,937 \times 4,005$ data matrix for clustering.

4 Analytical Framework

In this part of the study we turn our focus to grouping the extracted phase-locking patterns into a concise set of brain states. We apply a clustering pipeline introduced by Castro et al. (2024). We use k-means clustering to group these patterns into a small set of prototypical states. These centroids capture the essence of the brain's shifting dynamics, and serve as the basis for all subsequent analyses of state occupancy, transition dynamics, and their relationship to structural connectivity.

4.1 Convergence Analysis

Before setting our clustering parameters, we needed to ensure that our k-means solutions consistently arrived at the same centroids regardless of random initialization. This was essential not only for the stability of our results but also for the reproducibility of the entire analysis. The main parameter here is `n_init`, which controls how many times k-means restarts with different random seeds to find the best solution.

To evaluate convergence, we ran five independent clustering experiments ($n_{\text{exp}} = 5$) using fixed parameters ($k = 5$, `max_iter` = 40), while setting `n_init` = 20. Each row in Figure 3 shows the resulting centroids from one experiment, sorted by their similarity to the structural connectome (last column). This similarity is measured using the Structure–Function Correlation (SFC): the Pearson correlation between each centroid’s coherence matrix and the anatomical connectome. Higher SFC values indicate stronger alignment between a functional state and the brain’s underlying wiring.

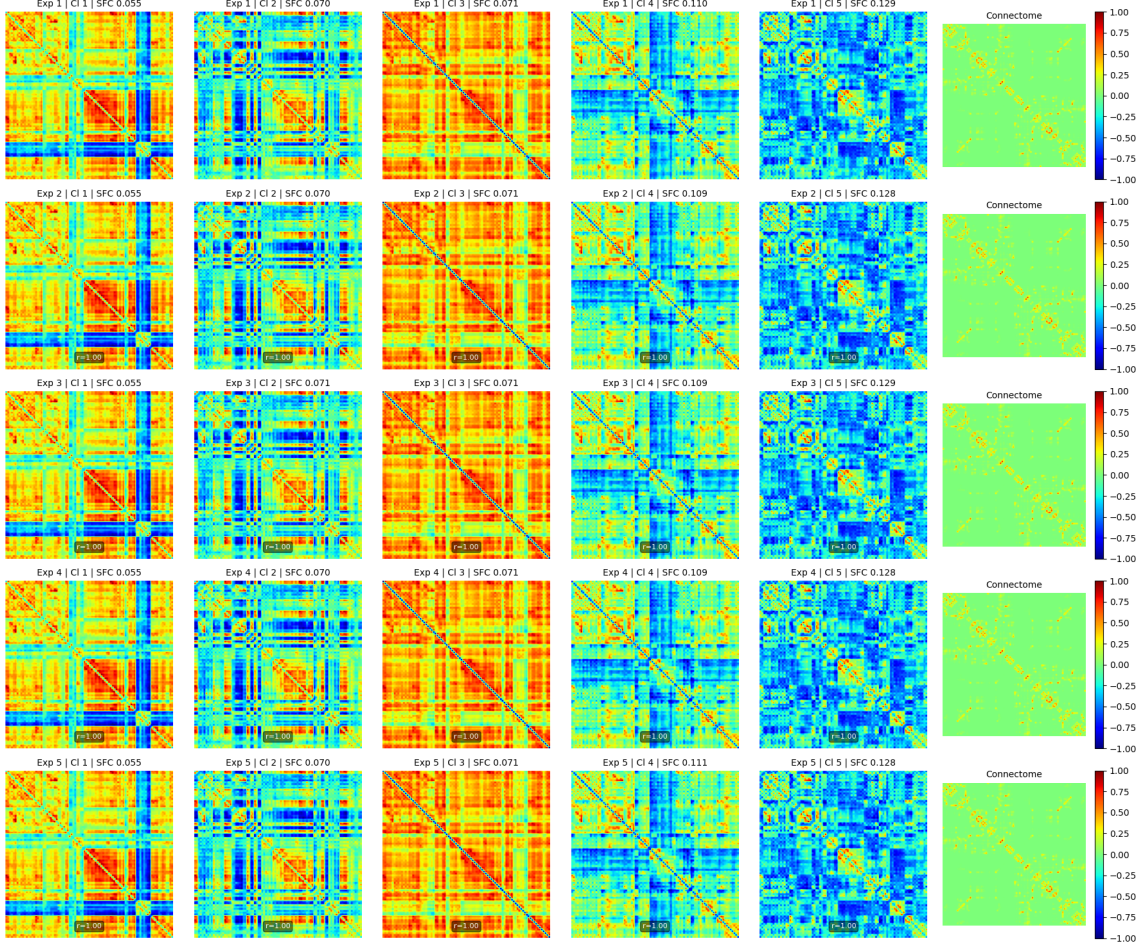


Figure 3: Convergence grid showing k-means clustering results across five independent runs ($n_{\text{exp}} = 5$) with `n_init` = 20. Each row represents one run. Columns 1–5 show the five phase-locking centroids, ordered by their correlation with the connectome (SFC score shown above each matrix). The last column displays the structural connectome used as reference. White text boxes indicate the Pearson correlation between corresponding centroids across experiments.

What we found is that once `n_init` reached 20, the centroids became highly consistent across runs. Correlations between matching centroids exceeded $r = 0.95$, and further increases in `n_init` did not improve this consistency. This analysis confirmed that our clustering pipeline converged reliably to a stable solution, and that our five canonical states are not artifacts of initialization, but reproducible features grounded in both functional and structural organization of the brain.

4.2 K-Means Clustering Setup

Following Castro et al. (2024), we set the number of clusters to $k = 5$, capturing the core repertoire of phase-locking states.

- **Distance metric:** Manhattan (ℓ_1) distance was used to reduce sensitivity to extreme phase-coherence outliers.

- **Initialization:** $n_{\text{init}} = 20$ random starts balance robustness against computation time.
- **Iterations:** $\text{max_iter} = 40$ ensures convergence without excessive looping.

4.3 Parameter Justification

The choice of $k = 5$ clusters derives directly from the validated pipeline of Castro et al. (2024), who used several number of clusters. Five centroids account for over 95% of the variance in principal eigenvectors extracted from phase-coherence matrices across a variety of altered conscious states.

Our convergence analysis (Sec. 3.2) confirmed that setting $n_{\text{init}}=20$ yields stable solutions: beyond this threshold, additional initializations provided negligible improvement in centroid reproducibility.

Finally, we opted for Manhattan (ℓ_1) distance rather than the more common Euclidean (ℓ_2) metric because ℓ_1 is less sensitive to extreme values and outliers in high-dimensional coherence patterns. By using ℓ_1 distance, we make the clustering more resilient to outliers, so that each group reflects true, recurring phase-locking patterns rather than brief bursts of noise.

4.4 Centroid Reconstruction and Structure–Function Alignment

Once k-means grouped our $6937 \times 4,005$ feature matrix into $k = 5$ clusters, each centroid $\mathbf{c}_m \in \mathbb{R}^{4005}$ captured a prototypical phase-locking pattern. For interpretation, we inverted the vectorization (using `squareform`) to recover each centroid’s full 90×90 coherence matrix.

We then aligned these functional patterns with anatomy by computing a Structure–Function Correlation (SFC) score:

$$\text{SFC}_m = \text{corr}(\text{vec}(\mathbf{C}_m), \text{vec}(\mathbf{C}_{\text{struct}})),$$

where $\mathbf{C}_m$ is the m th centroid matrix and $\mathbf{C}_{\text{struct}}$ is the anatomical connectome. Higher SFC indicates stronger alignment with white-matter wiring.

Finally, we ordered the five states by descending SFC, ensuring a consistent state labeling (State 1 = lowest SFC, $\dots$, State 5 = highest SFC) for all analyses of occupancy, transitions, and visualization (see Sec. 5).

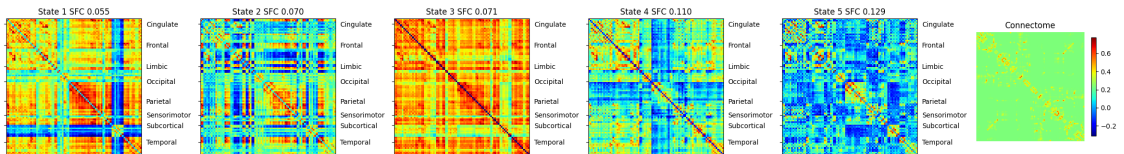


Figure 4: Visualization of the $k = 5$ centroid coherence matrices, sorted by their Structure–Function Correlation (SFC) score. Each panel displays a representative phase-locking pattern (functional state) derived from k-means clustering. Warmer colors indicate stronger phase synchrony between regions.

5 Key Findings and Insights

5.1 State Occupancy Analysis

We began by asking whether LSD shifts the brain’s preferred states away from the least anatomy-aligned patterns and toward others. Figure 5 (top row) displays the five centroids ordered by their Structure–Function Correlation (SFC) scores, from the weakest alignment (State 1, SFC = 0.055) through to the strongest (State 5, SFC = 0.129).

To see how much time the brain spends in each state, we plotted subject-level fractional occupancies under LSD (blue) versus placebo (red). We tested each pair of distributions for normality (Shapiro–Wilk) and then applied either a two-sample t -test or Mann–Whitney U test. In Figure 5, asterisks mark the states with significant differences at $\alpha = 0.10$.

Our results show:

- **State 1** (weakest SFC, 0.055): Occupancy drops under LSD, marginally significant ($p = 0.083$).
- **State 2**: (SFC = 0.070): No reliable change ($p = 0.792$)
- **State 3**: (SFC = 0.070): Slight increase under LSD, not significant ($p = 0.120$).
- **State 4** (SFC = 0.110): Occupancy rises under LSD, significant ($p = 0.032$).
- **State 5** (Strongest SFC, 0.129): Higher under LSD, not significant ($p = 0.139$).

At the group level (bottom row of Figure 5), bar plots confirm these trends: LSD reduces time in the weakest-linked State 1, boosts mid-range State 4 significantly, and shifts extra time into the most strongly anatomy-aligned State 5. Overall, these shifts indicate that LSD redistributes the brain’s time away from the least anatomy-aligned configurations and toward both intermediate and highly aligned states, reflecting a clear reorganization of conscious dynamics.

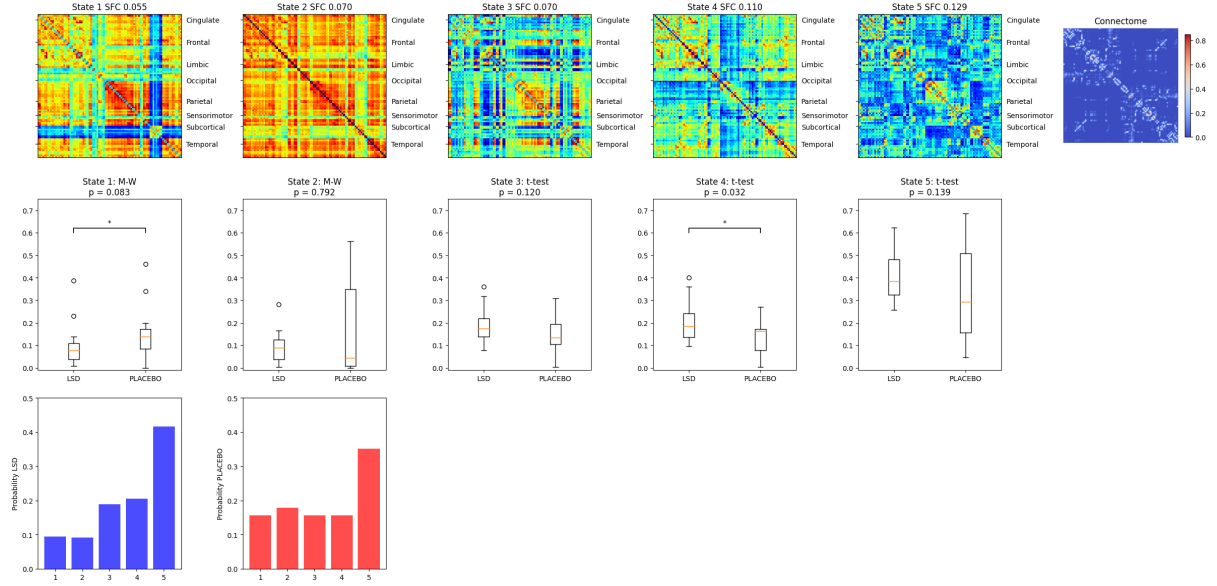


Figure 5: Top row: Coherence matrices for the five phase-locking states ordered by Structure–Function Correlation. Middle row: Boxplots of fractional occupancy under LSD (blue) and placebo (red); asterisks mark significant differences ($p < 0.10$). Bottom row: Group-level occupancy proportions, showing decreases in State 1 and increases in States 4 and 5 under LSD.

5.2 Shannon Entropy Analysis

After examining which states the brain prefers under LSD, we asked a broader question: *How evenly does the brain distribute its time across those states?*

To answer this, we computed the Shannon entropy of the state probabilities for each subject and condition. This measure captures the overall diversity, higher entropy means the brain samples states more uniformly (greater variety), while lower entropy implies that a few states dominate.

For each subject, we calculated the entropy H using the standard formula:

$$H = - \sum_{i=1}^5 p_i \log_2(p_i),$$

where p_i is the probability of being in state i , estimated from the frequency of state assignments across timepoints.

We then compared the entropy values between LSD and placebo using appropriate statistical tests: normality was assessed with the Shapiro–Wilk test, followed by either a paired t -test or a Wilcoxon signed-rank test.

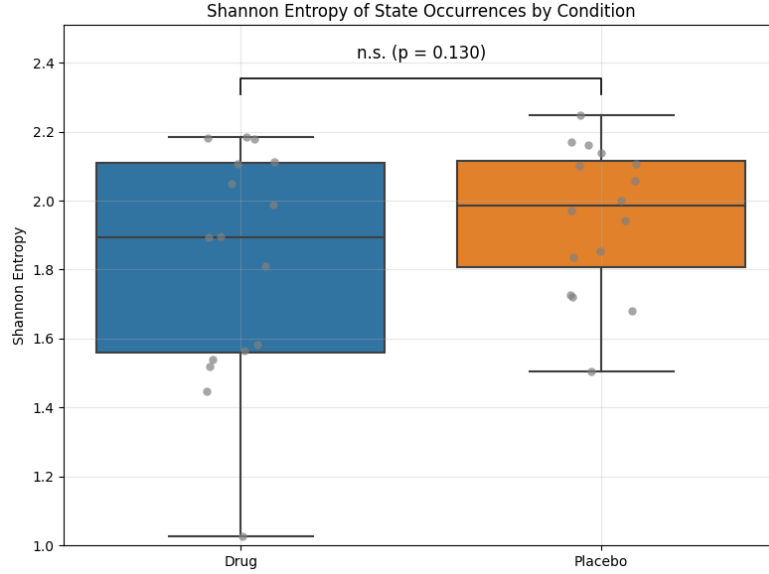


Figure 6: **Shannon Entropy of State Occurrences by Condition.** Each dot represents a subject’s entropy score under LSD (blue) and placebo (orange). Although LSD shows slightly more variability, the distributions do not differ significantly ($p = 0.130$).

As shown in Figure 6, the difference in entropy between LSD and placebo was *not statistically significant* ($p = 0.130$). This means that while LSD changes which brain states are visited more, the overall variety of states the brain uses stays about the same.

5.3 Markov Chain Entropy Analysis

While Shannon entropy captures the overall diversity of states, here to measure how unpredictably the brain jumps from one state to another, we built an Entropy Markov-chain.

We started by removing repeated states in each subject’s sequence so that staying in the same state wouldn’t dominate our counts. From this sequence, we counted how many times the brain moved from state m to state n , forming a raw transition matrix T . We then converted these counts into probabilities by normalizing each row of T , and finally used an iterative method to find the long-term (“stationary”) distribution π .

$$\pi P = \pi,$$

The invariant measure π gives the limiting probability of being in each state after many transitions, regardless of the starting state. In other words, π_i is the proportion of time the system spends in state i in the long run.

Finally, we quantified unpredictability with the Markov chain entropy

$$H = - \sum_i \pi_i \sum_j p_{ij} \log_2(p_{ij}).$$

- π_i is the invariant measure for state i (long-term probability of being in state i)
- p_{ij} is the transition probability from state i to state j

Higher entropy indicates more uncertainty or randomness in transitions, on the other hand lower entropy reflects more predictable switching patterns. We then compared subject-level entropy values between LSD (“Drug”) and placebo using a paired Wilcoxon ($\alpha = 0.05$) signed-rank test (chosen after checking normality). Figure 7 shows the resulting distributions.

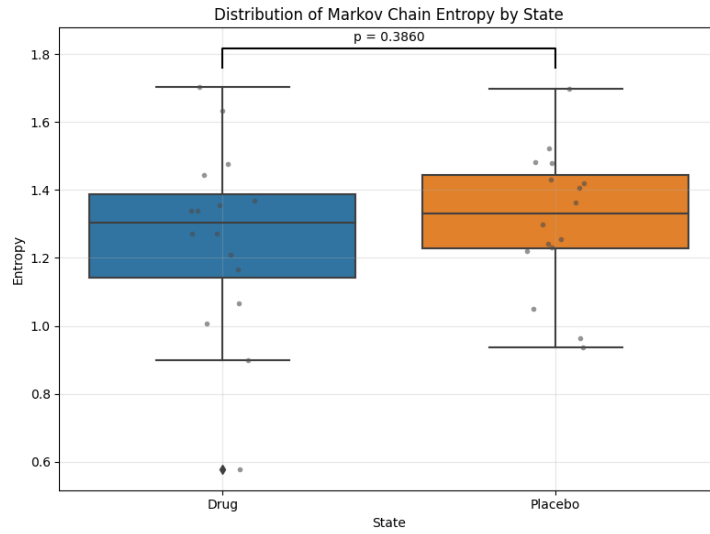


Figure 7: Distribution of Markov chain entropy under LSD (“Drug”, blue) and placebo (orange). Individual subject values are overlaid, and the Wilcoxon signed-rank test yields $p = 0.3860$, indicating no significant difference.

Contrary to our initial hypothesis, there was no significant change in transition entropy under LSD compared to placebo ($p = 0.3860$). Both conditions exhibit comparable levels of unpredictability in state switching, suggesting that while LSD reshapes which states are visited more, it does not substantially alter the overall randomness of transitions between them.

5.4 Brain Chord Diagram

To visualize how LSD reshapes large-scale network interactions, Figure 8 presents chord diagrams for the two states with significant occupancy shifts (State 1 and State 4). We aggregated the 90×90 coherence matrices into nine macro-areas (Cingulate, Frontal, Limbic, Occipital, Parietal, Sensorimotor, Subcortical, Temporal, Other) and plotted only the strongest links ($|r| > 0.7$).

- **State 1 (suppressed by LSD):** Very few robust connections, primarily Sensorimotor $\rightarrow$ Other, reflecting a modular, anatomy-bound configuration that LSD diminishes.
- **State 4 (enhanced by LSD):** Numerous strong chords among Frontal, Limbic and Sensorimotor areas, indicating heightened cross-network integration under LSD.

These diagrams illustrate LSD’s push away from segregated, low-SFC states toward more integrated, functionally rich configurations.

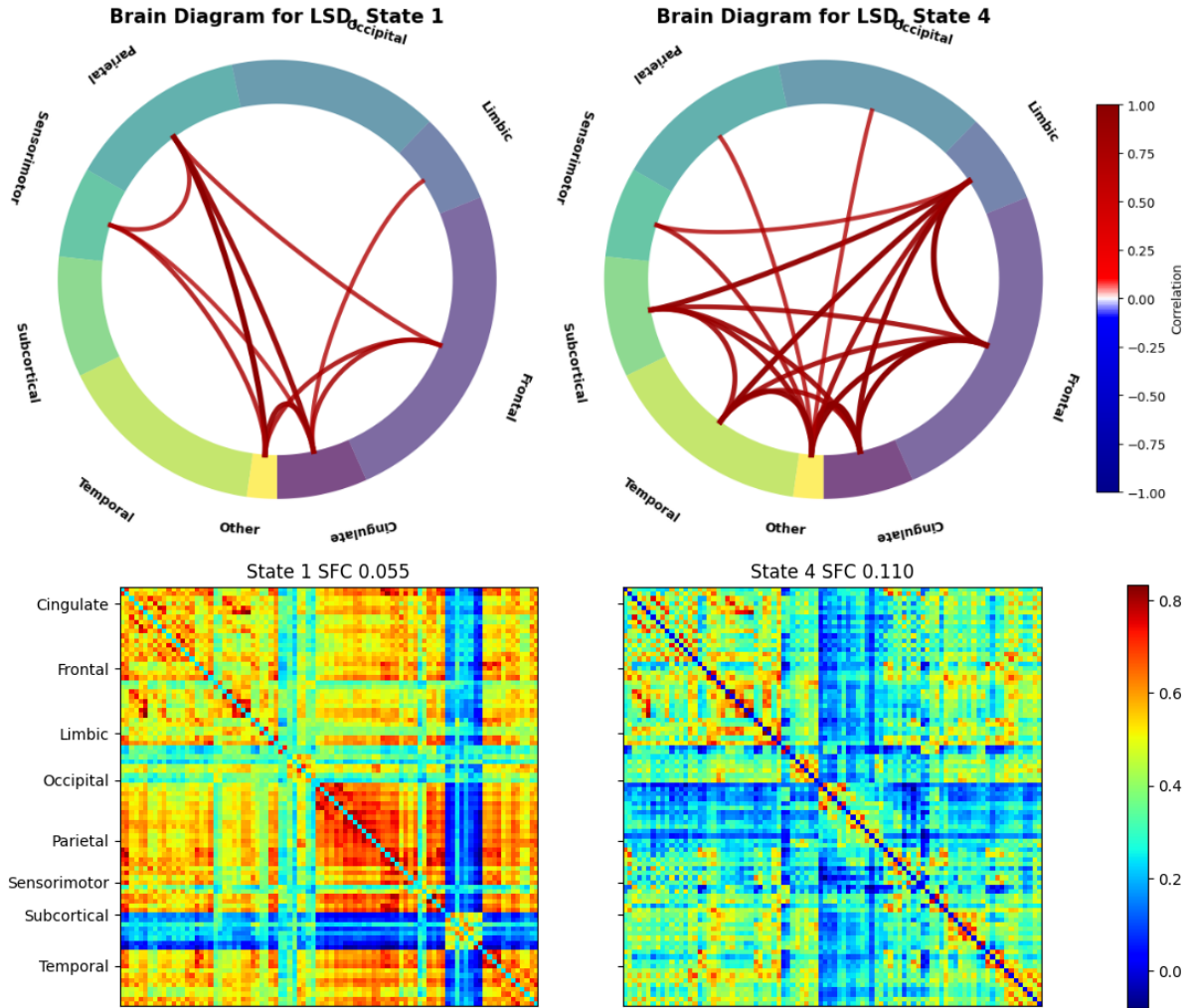


Figure 8: **Chord Diagrams for LSD-Responsive States.** (a) State 1: few strong links (mainly Sensorimotor–Other), the modular pattern reduced by LSD. (b) State 4: widespread Frontal–Limbic–Sensorimotor integration, the cross-network coupling boosted under LSD. Only correlations $|r| > 0.7$ are shown; arc lengths are proportional to macro-area size.

5.5 Subject-Level Correlation Matrices

As a final quality check and to illustrate inter-subject variability, we visualized each participant’s full 90×90 functional connectivity matrix under LSD and Placebo. Each panel displays the Pearson correlation between every pair of brain regions. All matrices use the same color scale

(-1,1) so patterns are directly comparable.

These grids let us verify that no subject's data are corrupted. They also provide context for our group-level findings by showcasing the range of individual brain architectures.

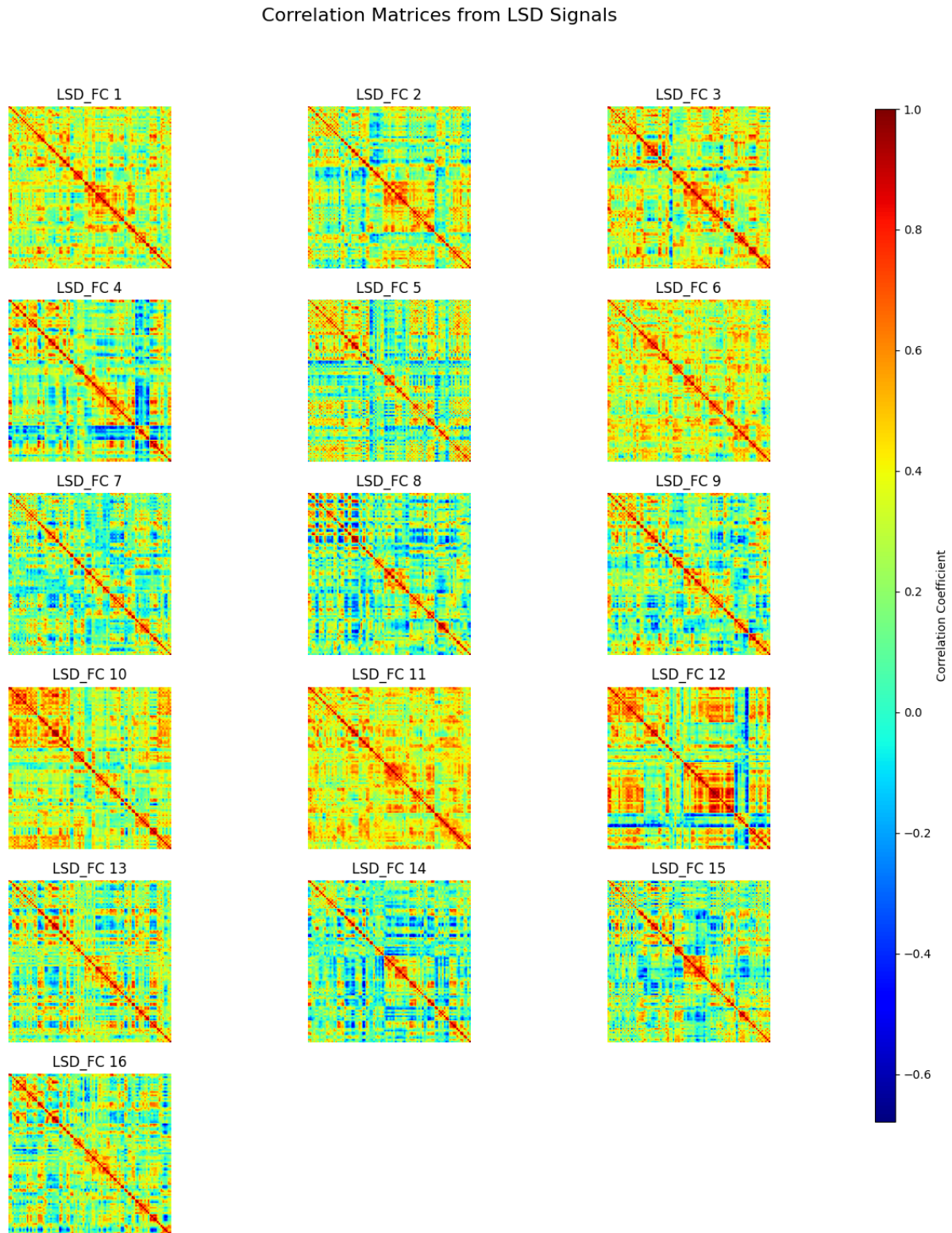


Figure 9: Functional connectivity matrices for each subject under LSD.

Correlation Matrices from Placebo Signals

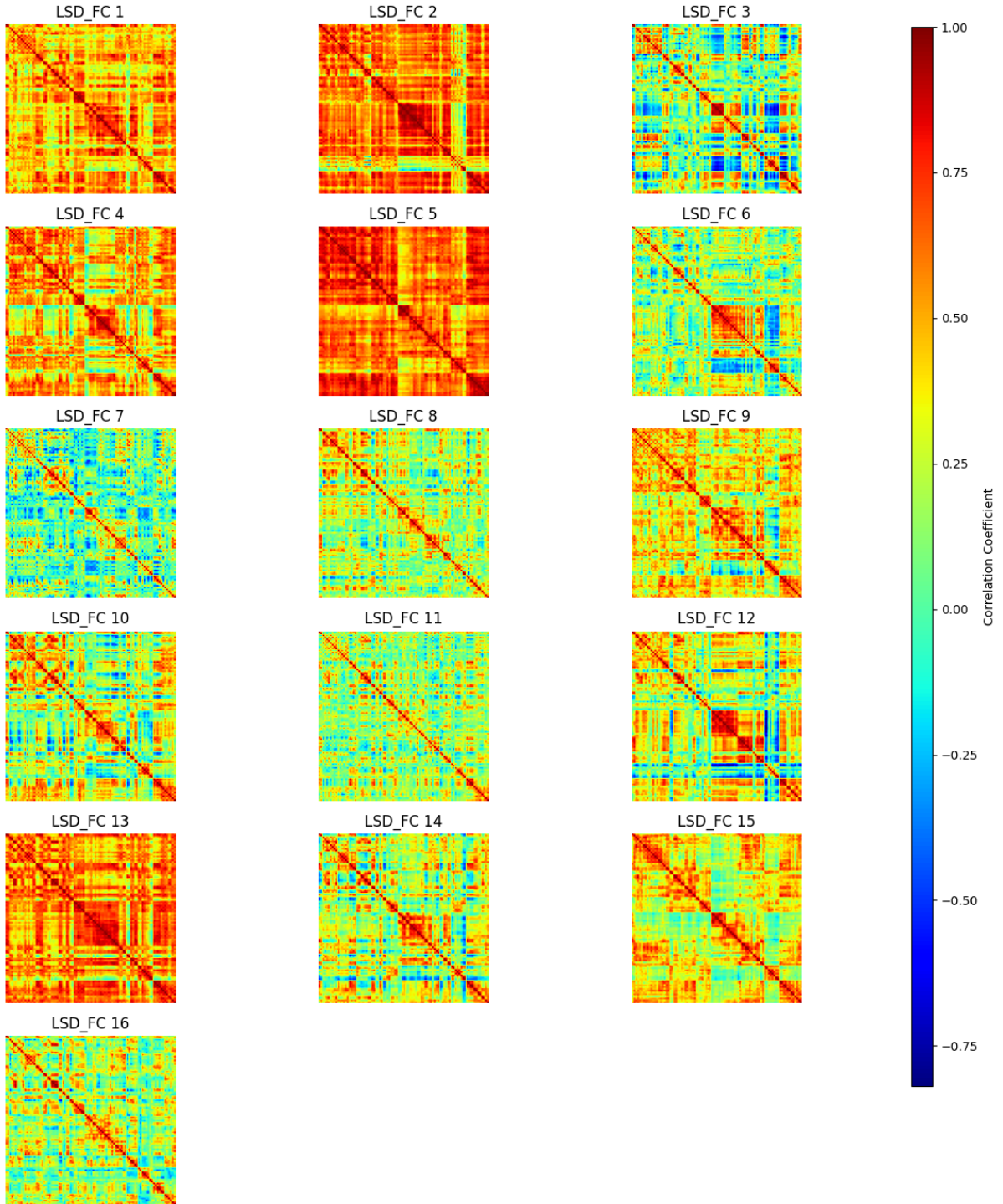


Figure 10: Functional connectivity matrices for each subject under placebo.

Across both conditions, all subjects exhibit the expected modular structure intraregion correlations along the diagonal, but LSD matrices (Fig. 9) often display enhanced off-diagonal coupling, especially between associative and limbic regions. In contrast, placebo matrices (Fig. 10) appear more locally constrained.

6 Discussion

Our results reveal that LSD reconfigures the brain’s dynamic repertoire rather than simply “adding noise.” Specifically:

- **State occupancy shifts:** Under LSD, time in the most anatomy-bound configuration (State 1) drops, while mid- and high-alignment states (notably State 4, and to a lesser extent States 2 and 5) become more prevalent.
- **Stable transition randomness:** Despite these occupancy changes, the overall unpredictability of state-to-state jumps (Markov-chain entropy) remains unchanged.
- **Flexible integration vs. segregation:** Chord diagrams show that LSD allows the brain to shift between almost global synchrony (State 3) and more specific connections between subsystems (e.g., Sensorimotor–Other in State 1, Frontal–Limbic in State 4).

By combining occupancy, entropy, and network visualizations, we find that psychedelics shift which brain states dominate while preserving the brain’s inherent flexibility. This dual effect, shifting how often each state occurs while keeping smooth transitions, might help explain how LSD can feel both “mind-expanding” and “deeply focused”.

6.1 Future Directions

This eight-week exploration opens several exciting pathways:

- **Cross-drug signatures:** Apply the same k-means pipeline to other substances: psilocybin, DMT, ketamine.
- **Longitudinal and clinical studies:** Follow participants over multiple dosing or therapeutic sessions to link state metrics (occupancy, SFC, entropy) with mood, cognition, or symptom trajectories.
- **Multi-modal integration:** Merge fMRI-based state analyses with EEG measures of signal diversity to capture fast and slow dynamics together.
- **Personalized parcellations:** Use subject-specific or higher-resolution atlases (e.g. Schaefer 400) to refine state definitions and detect subtler network shifts.

Pursuing these directions will deepen our mechanistic understanding of how psychedelics modulate the brain structure, dynamic state repertoire, and conscious experience.

7 Conclusion

In this study, we adapted a validated dynamic functional connectivity pipeline, centered on k-means clustering, to examine how LSD alters the brain's dynamics of phase-locking states.

We showed that LSD systematically shifts which states dominate, reducing time in the most anatomy-bound configuration while boosting mid- and high-alignment patterns, and yet preserves the overall unpredictability between states unchanged. Chord-diagram visualizations highlighted how LSD alternates the brain between widespread integration and focused modular interactions.

By linking these dynamic changes in structural anatomy (via SFC scores) and introducing clear visualization metrics, we provide a reproducible approach for studying the neural correlates of altered consciousness. Beyond refining theories of psychedelic action, our findings point toward practical, state-based biomarkers that could guide and monitor psychedelic-assisted therapies.

Code Availability

The full analysis pipeline (Python scripts and Jupyter notebooks) is publicly accessible at: https://github.com/DBlassio/phase_coherence_consciousness

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